

Delhi Public School

Subject: mathematics | Grade: 10th | Class: 10th

Time: 2 Hours

Maximum Marks: 30

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. Read all questions carefully before answering.
2. Write neatly and legibly.

Section A

Objective Type Questions (Multiple Choice Questions and True/False). Each question carries 1 mark.

Q1. Which of the following is a quadratic equation? [Easy] [1]

- a. $x^2 - 2x = (x - 1)^2$
- b. $x(x + 1) + 8 = (x + 2)(x - 2)$
- c. $x(2x + 3) = x^2 + 1$
- d. $x^2 + 3x = (x + 3)^2$

Q2. If one root of the quadratic equation $2x^2 + kx - 6 = 0$ is 2, then the value of k is: [Easy] [1]

- a. 1
- b. -1
- c. 2
- d. -2

Q3. What is the discriminant of the quadratic equation $3x^2 - 2x + 1/3 = 0$? [Moderate] [1]

- a. 0
- b. 1
- c. 2
- d. -3

Q4. If the roots of the equation $px^2 + qx + r = 0$ are real and equal, then which of the following relations is correct? [Moderate] [1]

- a. $q^2 = 4pr$
- b. $q^2 > 4pr$
- c. $q^2 < 4pr$
- d. $q^2 = pr$

Q5. If the equations $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ (where a is not equal to b) have a common root, then the value of (a + b) is: [Hard] [1]

- a. 1
- b. -1
- c. 0
- d. 2

Q6. State whether the following statement is True or False: Every quadratic equation has at least one real root. [Easy] [1]

Q7. State whether the following statement is True or False: If the coefficient of x^2 and the constant term of a quadratic equation have opposite signs, then the quadratic equation must have real roots. [Moderate] [1]

Section B

Short Answer Type Questions. Each question carries 2 marks. Show all steps clearly.

Q8. Find the roots of the quadratic equation $x^2 - 3x - 10 = 0$ by using the factorization method. [Easy] [2]

Q9. Find the values of k for which the quadratic equation $kx(x - 2) + 6 = 0$ has two equal real roots. [2]

[Moderate]

Q10. Solve the quadratic equation $2x^2 - 7x + 3 = 0$ using the quadratic formula. [Moderate] [2]

Q11. Find the roots of the equation $x + 1/x = 3$, where x is not equal to 0. [Hard] [2]

Section C

Long Answer Type Questions. Each question carries 3 marks. Draw diagrams and write detailed formulations where necessary.

Q12. Solve the equation for x : $1/(x + 4) - 1/(x - 7) = 11/30$, where x is not equal to -4 or 7 . [Easy] [3]

Q13. The diagonal of a rectangular field is 60 meters more than the shorter side. If the longer side is 30 meters more than the shorter side, find the sides of the field. [Moderate] [3]

Q14. Sum of the areas of two squares is 468 sq.m. If the difference of their perimeters is 24 m, find the sides of the two squares. [Moderate] **[3]**

Q15. Two water taps together can fill a tank in $9\frac{3}{8}$ hours. The tap of larger diameter takes 10 hours less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank. **[3]**

[Hard]

Q16. An express train takes 1 hour less than a passenger train to travel 132 km between Mysore and Bangalore. If the average speed of the express train is 11 km/h more than that of the passenger train, find the average speed of both trains. [Moderate] **[3]**

— End of Question Paper —